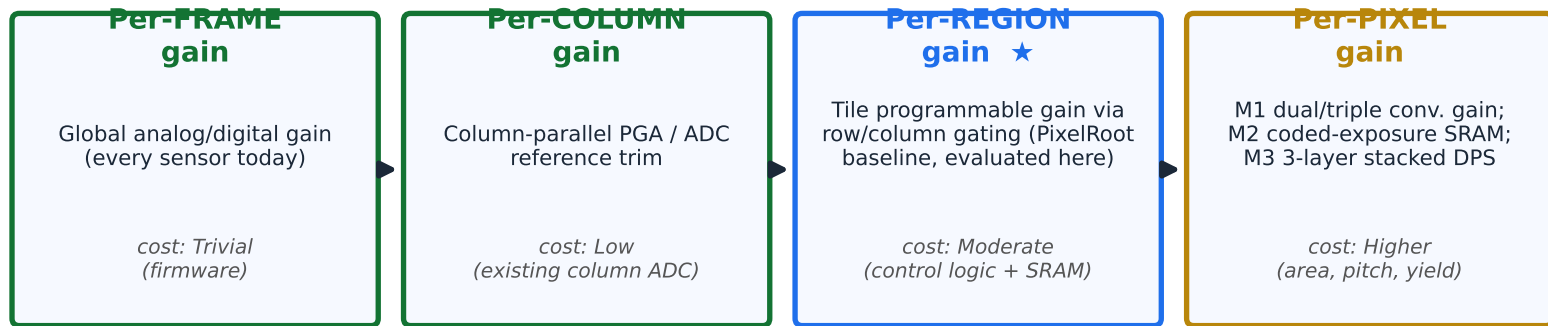


The provenance-write gain-control spectrum: granularity vs. manufacturing technique



increasing spatial resolution of the embedded mark →

increasing manufacturing investment →

PixelRoot's robust channel is realizable TODAY at per-region granularity; per-pixel is a manufacturable upgrade (M1 -M3), not a prerequisite.